Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/11/2025
PI Name:	Fanchiang, Christine Ph.D.		
Project Title:	HCAAM VNSCOR: Using a Human Capabilities Framework to Quantify Crew Task Performance in Human-Robotic Systems		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	650-302-2692
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City:	Centennial	**State:**	CO
Zip Code:	80122-1801	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	04/15/2019	**End Date:**	08/31/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2024 per NSSC information (Ed., 3/10/24) NOTE: End date changed to 4/14/2024 per A. Beitman/JSC (Ed., 2/20/23) NOTE: Start date changed to 4/15/2019 per NSSC information (Ed., 5/18/21) NOTE: End date changed to 4/14/2023 per NSSC information (Ed., 1/22/2020)		
Key Personnel Changes/Previous PI:	March 2020 report: Change to PhD graduate student for upcoming semester. August 2021 report: Added a MS graduate student (Kaitlyn Hauber) for the 2021-2022 academic year to help deploy subject testing July 2021 report: Added a Postdoc Dr. Katya Arquilla (volunteer) to help with data analysis May 2022 report: Removed a MS student (Kaitlyn Hauber) because she graduated July 2022 report: Postdoc upgraded to Professor (Dr. Katya Arquilla) at MIT Sept 2022 report: Added a PhD student (Amelia Gagnon) at MIT to help with HERA fNIRS data analysis because of her background in brain biosignals (EEG)		
COI Name (Institution):	Klaus, David Ph.D. (University of Colorado, Boulder) Shelhamer, Mark Sc.D. (Johns Hopkins University)		

Grant/Contract No.:	80NSSC19K0655
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This task is part of the Human Capabilities Assessments for Autonomous Missions (HCAAM) Virtual NASA Specialized Center of Research (VNSCOR). Effective space exploration will require proper task coordination between humans and robotic systems. These systems can be characterized in a variety of ways, from level of autonomy to the number of functions provided. At the most basic level a robotic system can be considered a hand tool while something more complex could be a humanoid companion. To ensure the robotic system is effective, the crew must trust that the system performs its intended function(s), or retain enough Situation Awareness (SA) and capability to find another way to execute the required task. Currently, there are no comprehensive standards for measuring, monitoring, and evaluating task performance with regard to crewmember capabilities, the design of the task, and the dynamic spacecraft environment. This work seeks to address this missing performance infrastructure by providing a conceptual framework for measuring task design quality and developing a path for validation using a task performance metric through experimentation both in university labs and using NASA's analog missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Leveraging wearable technologies for monitoring human health and performance is beneficial for a range of people here on Earth whether it is for elite athletes to rehabilitation patients in the hospital. Currently, there are limitations to the use and integration of data from various wearable sensors. The work done for this project will help to provide some guidance regarding wearable data integration and effectiveness of the data for predicting performance degradation. The ability to predict changes in performance can be useful for a number of scenarios here on Earth.
Task Progress:	As space exploration missions move further from Earth with limited communication capabilities, timely monitoring and assessment of crew health, status, and performance without the extensive level of ground support historically available in spaceflight operations becomes challenging and will necessitate a paradigm shift in crew monitoring. The objective of this investigation is to provide a method for validating a previously defined Capabilities Framework, which describes the relationship between human capabilities and performance. To approach this, our work focused on two complementary components: 1) the assessment of the framework for how we could quantify performance shaping factors using available commercial hardware and 2) collect the selected measures across simulated spaceflight environments and analyze whether these measures tell us anything insightful about the design quality and its influence on performance outcomes. The approach is to have human subjects perform representative spaceflight-like task scenarios while being monitored by a suite of non-invasive biometric measures. We ran three testing regimes, where each one investigated different aspects of the task design and its relationship to biosignals. In Block 1 we looked at the consistency of the biosignals over multiple days doing the same task, while in Block 2 we looked at whether we could distinguish biosignals between different task types, and in NASA's Human Exploration Research Analog (HERA) we investigated how the measures differ in a simulated spaceflight environment. The outcomes showed that biomeasures can be consistent across multiple days depending on the task type, but in such cases, the biomeaures do not provide any useful differentiation for comparing performance outcomes. But in cases where biomeasures are inconsistent, there is no clear correlation to performance or task design. There was an indication that in some cases, biomeasures can provide a slightly better chance at estimating the crewmember's workload state, but the low accuracy would currently not be useful for operations. However, we also found that using questionnaires also had lower predictive power for the subject's workload and performance levels than expected, which indicates that there is a need for more accurate measures to assess the state of the crewmember. While there is much more to investigate regarding how to best monitor and predict astronaut performance during real operations; the research performed for this grant has laid the groundwork for the potential of commercial off-the-shelf wearables as one method for predicting user workload and thereby providing a mechanism to better evaluate human-automation system designs.
Bibliography Type:	Description: (Last Updated: 02/26/2025)
Articles in Peer-reviewed Journals	Zero M, Klaus D, Arquilla K, Fanchiang C. "Defining and measuring crewmember operational state for spaceflight operations." Aerosp Med Hum Perform. 2024 Dec;95(12):919-29. https://www.doi.org/10.3357/AMHP.6405.2024 ; PubMed PMID: 39711412 , Dec-2024